



OPEN Multi-objective contextual bandits in recommendation systems for smart tourism

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In the context of smart tourism, recommender systems play a pivotal role in enhancing the personalization and quality of travel experiences. Tourists often face challenges in decision-making due to information overload. While context-aware recommender systems provide promising solutions by utilizing dynamic contextual data such as time, weather, and location, they struggle to adapt to real-time changes and to balance multiple objectives effectively. To address these challenges, this paper introduces a novel multi-objective contextual multi-armed bandit (MOC-MAB)-based recommender system. This approach integrates the strengths of contextual bandit algorithms with multi-objective optimization to provide personalized recommendations while simultaneously considering relevance and fairness. The proposed system dynamically learns from user feedback to optimize multi-objective recommendations. Extensive experiments conducted on a designed dataset simulating real-world scenarios and the TripAdvisor dataset demonstrate the approach's superior performance in terms of cumulative reward, click-through rate, and regret minimization when compared to baseline methods. This study also illustrates its practical application in the smart tourism context of Marrakesh, showcasing its potential to enhance tourism experiences in smart cities.

Keywords Multi-objective optimization, Contextual bandits, MAB, Recommender system, Smart tourism, Smart cities

Background and motivation

The emergence of smart cities has leveraged disruptive technologies and data-driven decision-making to enhance the quality of the citizens¹. Within rapidly developing smart cities, smart tourism has gained increasing attention as a means to provide personalized experiences to tourists by enhancing the travel industry^{2,3}. A prime example of a tourist destination that has embraced smart tourism is Marrakesh, a vibrant city known for its rich cultural heritage and diverse tourist attractions. For instance, through websites like “Hip Marrakech” and “Visit Marrakech”, tourists can navigate the city, find accommodations, make purchases, and make reservations through their smartphones, enhancing their overall experience⁴.

In the era of information overload, tourists face overwhelming choices when it comes to selecting accommodations, activities, and attractions. Recommender systems play an important role in alleviating this problem by providing personalized recommendations that match the user's preferences⁵. Thus, in the framework of smart cities, recommender systems become essential, they not only streamline the decision-making process for tourists but also maximize their engagement with the city. Utilizing recommender systems becomes particularly crucial by contributing significantly to the city's digital ecosystem⁶. By leveraging users contextual information, recommender systems can enhance the travel experience by ensuring that tourists make the most of their visit. As a sub-field of recommender systems, Context-aware Recommender Systems (CARSs) emphasize the incorporation of contextual factors, such as time, weather, and location, which serve to infer specific information that enhances personalized recommendations^{7,8}. CARSs extend analyzing user-item interactions by integrating various contextual factors to enhance the recommendation process. Thus, these systems offer distinct advantages over traditional algorithms. Traditional recommendation algorithms, such as collaborative filtering and content-based filtering, have limitations in incorporating contextual information, which can significantly impact the accuracy and relevance of the recommendations. This is particularly crucial in the context of smart tourism, where the success of the travel experience depends on providing personalized and relevant recommendations that align with the user's preference and context⁹. For instance, recommending outdoor activities on a rainy day or suggesting a restaurant that is closed during the user's preferred dining time can result in poor user experience and dissatisfaction.

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Research challenges

Despite the significant progress in CARS research, their algorithms often rely on static models that lack the ability to learn and adapt from ongoing user interactions. This limitation becomes more pronounced as user preferences and contextual factors evolve over time. The static nature of these systems can result in a degradation of performance, as they fail to capture the dynamic nature of user behavior and context. Hence, effectively capturing complex contextual information is challenging. The real-world context in smart tourism is multidimensional and dynamic, consisting of contextual factors such as time, location, weather, local events, and others. Therefore, it is essential for CARS to continuously adapt to these contextual changes in real-time. This necessitates the development of algorithms and techniques that can efficiently handle dynamic contextual information and make timely recommendations that align with the evolving needs of tourists.

To overcome these challenges, researchers have turned to reinforcement learning (RL) algorithms, which allow the system to learn from feedback by interacting with the environment^{10–12}. Hence, as item and user data is constantly updated, Recommender Systems (RSs) have transitioned from traditional batch learning methods to current online environments, thereby modeling these systems within the framework of the Multi-Armed Bandit (MAB) problem¹³. MAB problem models sequential decision-making, where it attempts to maximize the payoff from K slot machines, each with a different payoff probability. A natural extension of the k -armed bandit is the so-called contextual bandit, which involves user features and item features in deciding the best 'arm' or action¹⁴. Solutions involve algorithms like epsilon-greedy, upper confidence bound, Thompson sampling, and Linear Upper Confidence Bound (LinUCB) which aim to maximize total payoff or minimize regret relative to optimal strategy. The goal of contextual bandit algorithms used for recommendation systems is to recommend items to users based on their preferences and context¹⁵. Unlike traditional bandit algorithms that focus on exploring and exploiting different arms (recommendations) to maximize immediate rewards, contextual bandit algorithms take into account contextual information (user features, item features, and another relevant context) to make more personalized recommendations. Contextual bandits will enable the recommender system to provide information to users based on their context, and then learn and adapt from the feedback received. For example, a contextual bandit-based recommender system (contextual-BRS) can take into account the user's location, time of day, and preferences to recommend a nearby museum that matches their interests. Besides, contextual-BRS addresses the previously mentioned limitations. It provides a solution to the cold start problem by allowing exploration and exploitation even with limited initial data. This enables the system to adapt and improve recommendations as it interacts with users. Indeed, contextual-BRS is able to address the exploration-exploitation trade-off by leveraging contextual information to guide the exploration and exploitation processes. This ensures that recommendations are both tailored to the user's preferences and open to discovering new and potentially interesting options. Moreover, contextual bandit algorithms excel in scalability, as they can update recommendations and handle a large number of users and items without the significant computational overhead.

In smart tourism, it is crucial to consider the dual importance of relevance and fairness in recommendations^{16,17}. Relevance focuses on the user's perspective; it considers user-oriented factors shaping the overall travel experience that influence the possibility of repeat visits and contribute significantly to user satisfaction. Traditional recommender systems emphasize one single objective by focusing on the user's interests and satisfaction. Single-objective recommender systems often neglect the principle of fairness. In this context, fairness implies the equitable promotion and representation of diverse offerings from tourism sites and service providers. This includes not only the users seeking recommendations but also the overall tourism ecosystem by promoting the diverse offerings of the tourism destinations and service providers. Fairness in our context relates to ensuring all locations, regardless of popularity, get equal chances of being recommended. It ensures all tourism businesses get a fair chance to attract visitors and grow their revenue. Such offerings include a range of unique local cuisine, child-friendly attractions, adventure activities, and shopping spots, all contributing to the richness and diversity of the tourist experience. However, balancing multiple objectives poses a challenge for traditional recommender systems, as they typically focus on optimizing a single objective or fail to consider the trade-offs between different objectives.

Contributions of this study

To address these issues, this paper proposes a novel approach of multi-objective context-aware recommender systems in the context of smart tourism within smart cities. We propose a multi-objective contextual multi-armed bandit (MOC-MAB) based recommender system that combines the benefits of contextual bandit algorithms and multi-objective optimization. By integrating contextual information into the recommendation process, the algorithm can adapt and learn from user feedback considering multiple objectives simultaneously. Furthermore, the approach incorporates state-of-the-art techniques of scalarization to perform multi objectives. For example, our proposed approach can recommend a historical monument to a history enthusiast and a family-friendly activity for a user traveling with children. An essential advantage of the proposed MOC-MAB algorithm lies in its ability to leverage rich contextual information, such as user demographics, preferences, time, and location, to deliver highly personalized and engaging recommendations. This level of personalization is particularly crucial in the context of smart tourism, where understanding and addressing diverse user needs enhance the overall travel experience. By dynamically integrating contextual data into the recommendation process, the MOC-MAB algorithm surpasses traditional methods, which often overlook these critical factors. For instance, the algorithm can recommend outdoor activities to a user on a sunny day or suggest dining options based on a user's location and dietary preferences, significantly improving user satisfaction and engagement. This emphasis on contextualization allows the system to better adapt to user behaviors and dynamically changing scenarios in real-world environments.

The main novelties of our proposal are threefold: (1) the integration of multi-objective optimization with contextual bandit algorithms to address the inherent trade-offs between relevance and fairness in smart tourism

recommendations; (2) the development of an adaptive framework capable of handling dynamic, real-time user contexts and learning from evolving user feedback; and (3) the practical application of this method in enhancing tourism experiences in real-world scenarios such as Marrakesh, demonstrating its effectiveness in promoting a balanced and personalized tourism ecosystem. These contributions collectively advance the field of context-aware recommender systems, particularly in dynamic and multi-objective environments.

The remainder of this paper is organized as follows: Section “**Motivating scenario**” provides a motivating scenario of our proposed approach. Section “**Related work**” provides a comprehensive literature review, highlighting the state-of-the-art of multi-objective optimizations and reinforcement learning approaches. Section “**Contribution**” presents the methodology and framework used in this research, detailing the algorithms of the proposed multi-objective contextual multi-armed bandit (MOC-MAB) algorithm for recommendations in smart tourism. Section “**Experimental evaluation**” presents the experimental setup and the corresponding experimental results. Finally, Section “**Conclusion and future work**” concludes the paper with a summary of the contributions and directions for future research in the field of a multi-objective contextual bandits-based recommender system for smart tourism.

Motivating scenario

Considering an example of a single-objective recommender system that prioritizes revenue generation for tourism businesses over user preferences. In this scenario, the recommender system primarily recommends popular tourist attractions that are well-known places to maximize revenue. However, this approach disregards the preferences of individual tourists for unique and personalized experiences. As a result, the recommendation experience fails to cater to the diverse interests and needs of tourists. In the context of smart tourism, considering multiple objectives in recommendations is essential to introducing novelty and serendipity into the user's travel experience, enabling them to explore a wide range of attractions and activities. In a real-world scenario, suppose a tourist is exploring the vibrant city of Marrakesh, known for its rich cultural heritage and diverse attractions. Tourist expects the smart tourism recommendation system to take into account their specific interests and preferences. For instance, they may have a passion for exploring historical monuments, indulging in local culinary delights, and preferring to avoid adventure activities. The smart city recommendation system has a comprehensive list of tourist spots, each characterized by contextual information. This contextual information includes elements such as historical relevance, food rating, kid-friendliness, entertainment options, nature quotient, shopping opportunities, and adventure activities. In this context, our proposed MOC-MAB algorithm can provide personalized recommendations that align with the tourist's preferences and optimize multiple objectives simultaneously. For instance, it can assign higher weights to historical relevance and food rating, as they are crucial to the tourist's interests while assigning lower weights to other elements that may have less significance to the tourist. Figure 1 depicts a practical application of our proposed algorithm within a recommendation application aiming to enhance personalized experiences in smart tourism. The proposed MOC-MAB based recommender system recommends a touristic spot by integrating contextual information concerning both the tourist and the recommendation items. Subsequently, the tourist's feedback serves as a reward for the Multi-Objective Contextual MAB algorithm, thus completing the cycle of personalized recommendations. This illustrative scenario operates in three key steps: it first integrates (1) contextual information from both the tourist and the items to be recommended. Subsequently, it makes (2) a recommendation of a touristic location. Finally, it receives (3) feedback from the tourist, which serves as a reward mechanism for the MOC-MAB algorithm, thereby enhancing the quality and relevance of future recommendations. The tourist is subsequently able to have a virtual reality exploration of the suggested touristic place. The virtual reality visiting experience not only amplifies the recommendation process but also facilitates a more informed feedback provision, further refining the recommendation system.

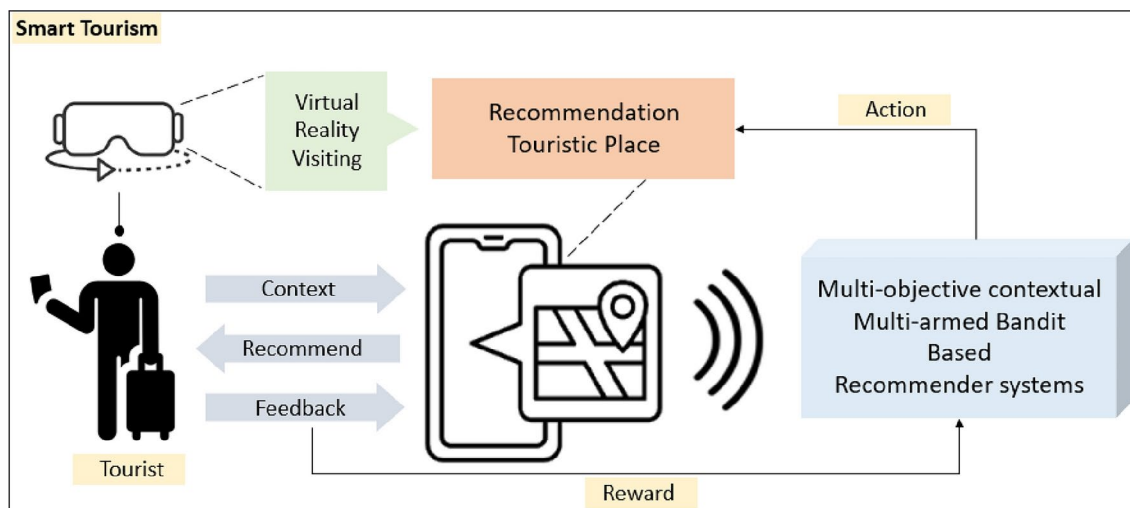


Fig. 1. Real-word recommendations for smart tourism.

Related work

The MAB problem, also known as the K-armed bandit problem^{18,19}, is a classic sequential decision-making problem that involves selecting from a fixed set of resources or arms to maximize expected rewards. The concept draws its name from a gambler at a row of slot machines in a casino, who seeks to maximize their profits by pulling levers and accumulating rewards. Each machine provides a random reward based on its own probability distribution, creating a dilemma for the gambler: should they exploit the machine with the highest expected payoff or explore other machines to gather more information²⁰? In the MAB framework, the learner sequentially selects arms and observes noisy reward samples from the chosen arms. The objective is to maximize the long-term reward by identifying arms with high rewards while avoiding excessive time spent on low-reward arms. This requires striking a delicate balance between exploration and exploitation, where the learner leverages the knowledge obtained through observations to make informed decisions. Ban et al.²¹ extend this concept by applying neural contextual bandits, which adapt to dynamic environments and effectively model complex user-item interactions.

Formally, let $A = \{a_1, a_2, \dots, a_K\}$ denote the set of K arms and $r_t(a_k)$ the reward obtained from pulling arm a_k at time t . The expected reward for arm a_k is $\mu_k = \mathbb{E}[r_t(a_k)]$. The goal is to maximize the cumulative reward over T iterations:

$$\text{Maximize } \sum_{t=1}^T r_t(a_t), \quad \text{where } a_t \in A. \quad (1)$$

The performance of an MAB algorithm is often evaluated using regret, defined as the difference between the reward of an optimal strategy and the cumulative reward of the algorithm:

$$\text{Regret}(T) = \sum_{t=1}^T r_t(a^*) - \sum_{t=1}^T r_t(a_t), \quad (2)$$

where $a^* = \arg \max_{a_k \in A} \mu_k$ is the optimal arm.

Mehrotra and McInerney²² highlight the importance of integrating fairness and diversity in bandit-based systems, which can mitigate biases in cumulative reward estimation.

Numerous MAB algorithms have been introduced in the literature, each with its distinct characteristics and properties^{23,24}. Among the prominent algorithms are the ϵ -Greedy approach²⁵, Upper Confidence Bounds (UCB) method^{26,27}, and Thompson Sampling (TS) technique²⁸. These algorithms offer different strategies for balancing exploration and exploitation in the decision-making process, contributing to the advancement of MAB solutions. Balakrishnan et al.²⁹ proposes an online system based on contextual bandits that learn behavioral constraints and incorporate them into real-time decision-making. The system identifies relevant context features aligned with the constraints and maintains performance while operating within them. In³⁰, an ILP-based algorithm called DPSS and a MAB-based algorithm called DPSS-UCB are introduced. These algorithms ensure that a subset of agents meets quality constraints, with DPSS handling known qualities and DPSS-UCB addressing unknown qualities. Sanz-Cruzado et al.³¹ present a multi-armed bandit extension of neighbor-based collaborative filtering, incorporating controlled stochastic exploration using Thompson sampling. A combinatorial multi-armed bandit algorithm called GLS-MAB is introduced in³². It learns uncertainties and determines consumer costs through an auction. The mechanism is regret optimal, incentive compatible, and demonstrated to be effective through simulations. Chen and Zhou³³ extend these principles by proposing an extended trust and distrust network-based dual fuzzy recommendation model (ETD-DFR). This model addresses fuzziness and uncertainty in user preferences, introducing a dual fuzzy framework and leveraging social balance theory and trust/distrust propagation for better recommendations. The approach uses the Louvain algorithm for efficient subgroup identification, ensuring precise and robust recommendations.

Chiang and Mohler³⁴ propose a framework that integrates Hawkes processes with multi-armed bandit algorithms for spatio-temporal event forecasting and detection. The objective of the corresponding algorithms is to maximize the overall reward while operating within a limited budget allocated to the decision-maker. The study by Zhang et al.³⁵ proposes a combined weighting-based large-scale group decision-making (LSGDM) framework for MOOC group recommendations. This approach leverages probabilistic linguistic term sets and word embeddings to integrate subjective and objective weighting, addressing challenges such as information overload and low course completion rates in MOOCs. The framework's application demonstrates its robustness and effectiveness in enhancing recommendation outcomes, making it relevant for systems aiming to handle complex user preferences in group decision-making scenarios.

By integrating context information into the decision-making process, contextual bandits offer an extension to the traditional MAB problem. This augmentation enables the algorithm to take into account contextual factors, such as user demographics, historical preferences, time, and location, among others. MOC-MAB algorithms excel in leveraging contextual information such as user demographics, preferences, time, and location to deliver highly personalized recommendations. These algorithms go beyond traditional methods by dynamically integrating real-time contextual data, enabling them to generate more relevant and engaging recommendations. For example, MOC-MAB algorithms can suggest a nearby attraction suited to a user's preferences and current location or recommend dining options considering the user's dietary restrictions and time constraints. This personalized approach significantly enhances user satisfaction and engagement, particularly in dynamic domains like smart tourism, where the interplay of various contextual factors is critical to optimizing the recommendation process. Taking advantage of contextual information, contextual bandits bridge the gap

between traditional MAB approaches and the need for more personalized and context-aware decision-making in recommendation systems³⁶.

Chen et al.³⁷ proposed the PLDTN-UCF model, which combines probabilistic linguistic representations with dual trust networks to enhance collaborative filtering recommendations. By transforming online user reviews into probabilistic linguistic ratings, the model captures the uncertainty in user opinions and attitudes. It further integrates multi-source interpersonal trust through an innovative dual trust network based on online behaviors and probabilistic linguistic ratings. Experimental evaluations using the Douban movie dataset demonstrated that PLDTN-UCF outperforms state-of-the-art models in accuracy and trust representation, highlighting its relevance in improving personalized recommendations. In³⁸, the DC3MAB algorithm is proposed to tackle the challenges of recommender systems in large-scale user-item scenarios with sparse interactions. It incorporates dynamic user clustering, collaborative filtering-based item partitioning, and a multi-class reward mechanism to improve preference mining, handles growing item sets, and capture rich implicit feedback. The challenge of generating successful reciprocal recommendations in user-to-user recommender systems is formulated as a contextual bandit problem³⁹. Chen et al.⁴⁰ propose a Dynamic Trust Network-Based Fuzzy Group Recommendation (DTN-FGR) model tailored for collective tourist destination recommendations. The model transforms user-generated ratings into fuzzy preference relations (FPR) to address inconsistent evaluation criteria among group members. Furthermore, it employs a PageRank-based method for calculating trust scores in a dynamic trust network and adapts to changing social relations. A case study demonstrates the model's reliability, stability, and superior consensus degree compared to other collective recommendation models.

Kumari et al.⁴¹ combines deep learning with contextual bandits to provide post-hoc argumentation-based explanations for reciprocal recommender systems (RRS). The cold-start problem, where traditional recommendation systems lack sufficient user data, is highlighted in⁴². A contextual bandit approach-based recommendation system for Personalized web-based services is proposed and evaluated using regret evaluation.

However, while contextual bandit algorithms excel in incorporating contextual information for optimizing a single objective, such as maximizing click-through rates or user engagement, they often do not explicitly address the challenges associated with multiple conflicting objectives. Multi-objective optimization aims to find a set of solutions that optimally balances multiple conflicting objectives simultaneously. By incorporating multi-objective optimization into contextual bandit algorithms, it becomes possible to consider and optimize for diverse objectives such as relevance, diversity, fairness, novelty, and serendipity in the recommendation process.

Techniques such as scalarization and Pareto optimization have been widely explored in the literature for multi-objective decision-making⁴³. These are further elaborated in the context of our proposed approach in Section "Proposed approach".

The CMAB-DO algorithm proposed in⁴⁴ addresses the contextual multi-armed bandit problem with a dominant objective. It introduces the MOC-MAB algorithm, which aims to maximize rewards in both dominant and non-dominant objectives. The algorithm achieves sublinear regret and demonstrates strong performance in designed and real-world datasets, making it applicable to various domains with conflicting objectives. In⁴⁵, an online gradient ascent learning algorithm is presented for optimizing multiple objectives in recommender systems used in multi-stakeholder platforms. The algorithm utilizes the Generalized Gini index to jointly optimize objectives and maximize long-term vectorial rewards. By considering multiple objectives, it aims to provide superior recommendations compared to alternative approaches across diverse scenarios. In⁴⁶, the focus is on multiple conflicting objectives where the optimal choice varies based on the context. The contextual Pareto regret is introduced as a performance metric, and the Pareto Contextual Zooming (PCZ) algorithm is developed. PCZ partitions the joint context-arm set to identify arms that are close to the Pareto front, enabling effective decision-making in complex multi-objective environments.

Table 1 summarizes the objectives, contextual integration, and key features of key methods from the literature compared to the proposed approach. Traditional methods like e-Greedy, UCB, and Thompson Sampling optimize single objectives but lack contextual integration and multi-objective capabilities. Contextual bandits such as LinUCB introduce contextual adaptation but remain focused on optimizing single objectives, typically click-through rates (CTR). Advanced approaches like CMAB-DO and GLS-MAB extend to multi-objective optimization but are primarily validated in static or theoretical settings.

The proposed MOC-MAB method stands out by integrating multi-objective optimization with contextual bandits, addressing conflicting objectives such as relevance and fairness. Furthermore, its application for

Method	Objective(s)	Contextual integration	Key features
e-Greedy ²⁵	Single Objective (Reward Maximization)	No	Simple exploration-exploitation mechanism for static environments
UCB ²⁶	Single Objective (Reward Maximization)	No	Optimizes rewards using confidence bounds but lacks contextual adaptation
Thompson Sampling ²⁸	Single Objective (Reward Maximization)	No	Bayesian inference for reward maximization; no multi-objective capability
LinUCB ⁴⁷	Single Objective (CTR Maximization)	Yes	Incorporates linear contextual information for personalized recommendations
CMAB-DO ⁴⁴	Multi-Objective Optimization (Dominant/Non-Dominant Objectives)	Yes	Scalarization-based optimization for conflicting objectives
GLS-MAB ³²	Regret Minimization	Partial	Combines contextual and combinatorial optimization for decision-making
Proposed Method	Multi-Objective Optimization (Relevance and Fairness)	Yes	Dynamically balances multiple objectives in real-world, dynamic smart tourism scenarios

Table 1. Comparison of existing methods and proposed approach.

recommending in real-world dynamic scenarios, such as smart tourism, demonstrates its effectiveness in balancing these objectives while adapting to user feedback in real-time.

To the best of our knowledge, no prior work has investigated the application of recommender system based on multi-objective contextual bandits in the domain of smart tourism. This study seeks to fill this research gap by introducing a novel multi-objective contextual bandit algorithm specifically designed for smart tourism applications. The algorithm's performance is evaluated in a real-world smart tourism scenario, providing valuable insights into its effectiveness and applicability in this domain.

Building on the strengths and limitations of existing methods, our proposed MOC-MAB algorithm uniquely integrates multi-objective optimization and contextual bandits, achieving a balanced trade-off between relevance and fairness. Unlike traditional single-objective approaches such as LinUCB, which primarily optimize immediate rewards or click-through rates, our algorithm incorporates scalarization techniques to effectively manage conflicting objectives. Furthermore, by dynamically exploiting contextual data, the MOC-MAB method adapts to real-time changes in user preferences and environmental conditions, thus addressing the shortcomings of static models highlighted in the literature. For example, although the CMAB-DO method incorporates multi-objective capabilities, it is validated mainly in static or simulated environments. In contrast, our algorithm demonstrates superior adaptability and performance in real, dynamic contexts such as smart tourism. This practical applicability is evident from the improvements observed in cumulative rewards and click-through rates in various datasets, as described in the experimental evaluation section.

Contribution

Problem formulation

Figure 2 illustrates the functioning of the multi-objective contextual multi-arm bandit recommendations through an iterative, four-step process over a series of users. (1) The agent observes a tourist (user) along with any available and pertinent context. (2) An action (i.e., arm, option, or touristic place) is assigned to the tourist, based on the obtained context and the agent's current internal model. (3) The chosen action is implemented and recommended to the tourist. (4) The tourist's response to the recommended action serves as the reward, which is then recorded by the agent. Using the gathered information; the user's reaction, the context, and the assigned action, the agent refines its approach for future interactions with similar users.

In the formulation of the multi-objective contextual bandit problem, let's denote a set of tourists as $T = \{t_1, t_2, \dots, t_N\}$, each with a unique set of preferences and objectives. For each time step t , the algorithm receives a context vector $x_t \in \mathbb{R}^d$ which can include various information about the tourist, such as location, time of day, weather conditions, and historical interactions. The algorithm then selects an action $a_t \in A = \{a_1, a_2, \dots, a_K\}$ (recommendation), where A is the set of possible actions. The goal is to choose an action that maximizes the cumulative reward $R_t(a_t)$ across all objectives $O = \{o_1, o_2, \dots, o_M\}$.

$$R_t(a_t) = \sum_{o_i \in O} r_{t,o_i}(a_t) \quad (3)$$

Where $r_{t,o_i}(a_t)$ is the reward associated with action a_t under objective o_i at time t .

After the action a_t is selected, the algorithm receives feedback $y_{t,o_i}(a_t) \in \mathbb{R}$ from the tourist, indicating the satisfaction or relevance of the recommendation under each objective o_i . The overall goal is to learn a policy

$\pi : \mathbb{R}^d \rightarrow A$ that can adaptively make recommendations by exploiting the observed context x_t while exploring new options. This policy should optimize the total cumulative reward across all time steps and objectives:

$$\sum_{t=1}^T \sum_{o_i \in O} y_{t,o_i}(a_t) \quad (4)$$

Where T is the total number of time steps.

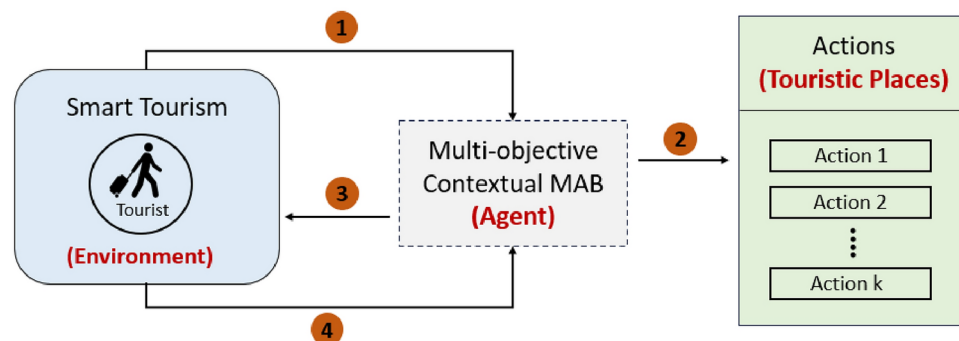


Fig. 2. MOC-MAB based recommendations.

The main challenge of this problem lies in balancing the trade-off between exploration and exploitation across multiple objectives, while effectively incorporating the dynamic feedback from the tourists.

Proposed approach

Our proposed MOC-MAB algorithm learns from the interactions and the associated rewards, while also taking into account the context vectors to make informed and personalized recommendations. It incorporates multi-objective optimization techniques into contextual bandit recommender systems. The multi-objective optimization allows for the simultaneous optimization of multiple conflicting objectives, enabling the system to provide recommendations that balance user satisfaction and business revenue. For example, the system can recommend a mix of popular and niche tourist attractions, ensuring a diverse range of options that cater to different user preferences and enrich the overall travel experience.

The overall framework of the proposed algorithm is depicted in Fig. 3. This figure provides a concise overview of the main components of the system, including the input data (tourist preferences and contextual information), the algorithmic process (contextual analysis, multi-objective scalarization, and action selection), and the output (recommended items and user feedback). This visual summary enhances the understanding of the method and its iterative workflow.

The proposed algorithm 1 leverages context vectors associated with each arm (touristic place) to assess the relevance and attractiveness of different touristic places, enabling it to tailor recommendations that align with the tourist's specific interests. Suppose a tourist is presented with n different touristic places as potential recommendations. Out of these n options, one place is randomly chosen as the recommended arm for that particular interaction. The tourist will provide a reward value, which indicates whether the tourist liked or did not like the recommended place. This reward serves as feedback for the algorithm to learn and improve its future recommendations. Our proposed algorithm leverages dimensional context vectors associated with each arm (touristic place) to assess the relevance and attractiveness of different touristic places, enabling it to tailor recommendations that align with the tourist's specific interests. Let $P = \{p_1, p_2, \dots, p_n\}$ be the set of touristic places, r_t be the reward received from the tourist at time step t , and $C = \{c_1, c_2, \dots, c_n\}$ be the set of associated context vectors.

The algorithm's goal is to select the recommended touristic place p_i that maximizes the expected reward given the context vector c_i . This can be expressed as:

$$\operatorname{argmax}_i \sum_{t=1}^T r_t \cdot \mathbb{I}(p_i = \text{recommended at time } t) \quad (5)$$

Where T is the total number of interactions and $\mathbb{I}(p_i = \text{recommended at time } t)$ is an indicator function that equals 1 if p_i is recommended at time t and 0 otherwise.

In this formula, the algorithm aims to maximize the cumulative reward obtained from the recommended touristic places, considering the feedback provided by the tourist. The context vectors associated with each touristic place contribute to the evaluation of the recommendations. In an online learning scenario, the proposed approach will typically interact with the environment or users in a sequential manner, making recommendations and receiving feedback in real-time. The learning process will occur incrementally as new data becomes available, allowing the algorithm to continuously adapt and improve its recommendations. Our multi-objective contextual bandit algorithm is an enhanced version of the LinUCB algorithm. Thus, the main objective in the LinUCB algorithm is to choose the arm/action that maximizes the upper confidence bound (UCB) of the estimated reward. This is done by considering both the expected reward and the uncertainty of the estimate. The expected reward is estimated through a linear function of the context, and the uncertainty is quantified through the variance of the estimated reward. It treats each context feature equally. In contrast, our proposed algorithm takes into account weights for the context features. These weights are multiplied by the corresponding context

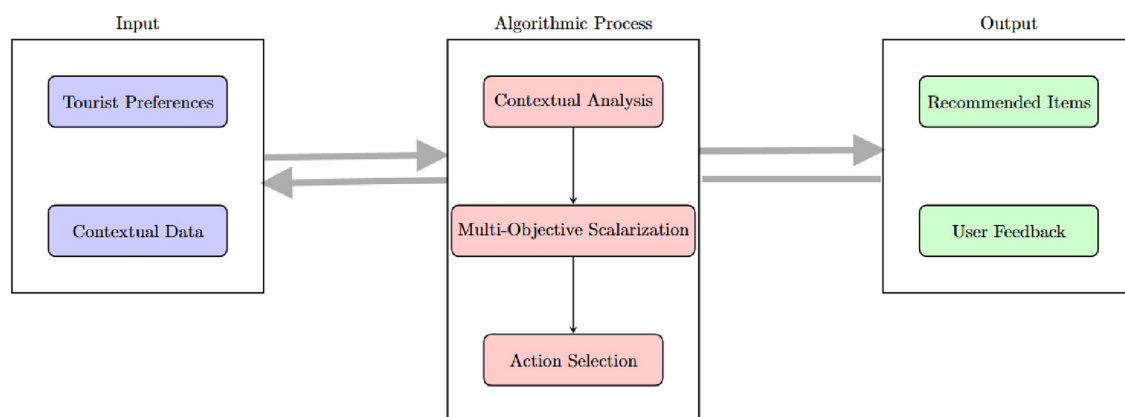


Figure 3. Framework of the proposed MOC-MAB algorithm.

features. This means that the proposed MOC-MAB algorithm uses a weighted linear function of the context to estimate the expected reward. This could be beneficial if some context features are known to be more relevant or important than others for predicting the reward. The weights can be learned based on prior experiences or domain knowledge.

Algorithm 1 takes in several parameters: the number of arms (possible actions) n_{arms} , the dimension of the context context_dim , a confidence level α , the data to process, the number of rounds to iterate, a flag train_flag that indicates whether to train the model and weights for each objective. Then, the algorithm enters a loop over each round in the dataset. At each round, the probability p_{ta} of taking each action (arm) is calculated based on the context, weights, and a multi-objective scalarization function. This function is designed to transform multiple objectives into a single metric that can be optimized. The algorithm selects the action with the highest p_{ta} , which represents the best trade-off among multiple objectives. Once the action is selected, the matrices A and b , as well as cumulative metrics, and their histories, are updated based on the chosen action and the reward obtained. The algorithm keeps track of cumulative metrics, such as cumulative reward and cumulative regret. Cumulative reward measures the total accumulated reward over time, indicating the overall effectiveness of the recommendations. Cumulative regret measures the accumulated difference between the reward of the recommended touristic place and the reward of the best possible choice, reflecting the algorithm's learning progress.

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1 Function MOC.MAB( $n, d, \alpha, \mathcal{D}, T, \text{train\_flag}, w$ ):
2    $A_i \leftarrow I_d, b_i \leftarrow 0_d$  for  $i = 1, \dots, n$ 
3    $R_c \leftarrow 0$ 
4   // Initialize cumulative reward
5    $\mathcal{R}_t \leftarrow \emptyset, \mathcal{L}_t \leftarrow \emptyset$ 
6   // Initialize reward and regret histories
7    $L_c \leftarrow 0$ 
8   // Initialize cumulative regret
9   for  $t \in 1, \dots, T$  do
10    for  $i \in 1, \dots, n$  do
11       $p_{t,i} \leftarrow w^T x_i + \alpha \sqrt{x_i^T A_i^{-1} x_i}$ 
12    end
13     $a_t \leftarrow \arg \max_i p_{t,i}$ 
14    // Select action with maximum score
15    Observe reward  $r_t$  and update  $R_c \leftarrow R_c + r_t$ 
16    if  $\text{train\_flag}$  then
17       $A_{a_t} \leftarrow A_{a_t} + x_{a_t} x_{a_t}^T$ 
18       $b_{a_t} \leftarrow b_{a_t} + r_t x_{a_t}$ 
19    end
20     $L_c \leftarrow L_c + \max_i p_{t,i} - r_t$ 
21    // Compute cumulative regret
22    Append  $R_c$  and  $L_c$  to  $\mathcal{R}_t$  and  $\mathcal{L}_t$ , respectively
23  end
24  return  $\mathcal{R}_t, \mathcal{L}_t$ 

```

Algorithm 1. Multi-Objective Contextual MAB (MOC-MAB)

Scalarization techniques for multi-objective optimization

Scalarization and population-based heuristics are widely used techniques in multi-objective optimization and decision-making⁴³. Scalarization is employed to convert a multi-objective problem (MOP) into a single-objective problem (SOP), allowing the utilization of existing optimization methods developed for SOPs using methods such as weighting methods⁴⁸, lexicographic⁴⁹, e-constraint⁵⁰, or Tchebycheff methods⁵¹, etc. This transformation simplifies the problem by assigning weights or preferences to the objectives, enabling a trade-off between them based on their relative importance. By solving the transformed SOP, a compromise solution that balances the multiple objectives can be obtained. Scalarization methods are popular due to their simplicity in converting MOPs to SOPs, enabling the use of existing optimization techniques for single-objective problems. Zhang and Li⁵² propose fairness-aware scalarization frameworks, effectively managing the trade-offs between multiple objectives such as fairness, relevance, and diversity in real-world applications. Additionally, scalarization methods often have lower computational complexity compared to population-based heuristics. They require solving a single-objective problem rather than maintaining and evolving a population of solutions. This advantage can be beneficial when dealing with computationally intensive or large-scale optimization problems. Scalarization techniques, as discussed above, are crucial for balancing the conflicting objectives of fairness and relevance in real-world smart tourism scenarios, ensuring a diverse yet personalized recommendation experience.

This algorithm represents an instance of a Multi-Objective Contextual MAB problem, in which actions are selected not just based on their expected immediate rewards, but also considering the multiple objectives of the recommendation scenario. Multiplicative scalarization is one approach to combining multiple objectives into a single value. Given n objectives, let $f_i(x)$ denote the i th objective function, where x is the solution vector, and let w_i denote the i th weight associated with that objective:

$$F(x) = \prod_{i=1}^n [w_i \cdot f_i(x)] \quad (6)$$

$F(x)$ is the result of the multiplicative scalarization, and it provides a single value that can be used for the comparison of different solutions. Note that w_i values are typically normalized so that their sum equals 1. Given n objectives with objective functions $f_i(x)$ and corresponding weights w_i , the objective is to find the solution vector x that maximizes the product of the weighted objective functions:

$$\max_x F(x) = \max_x \prod_{i=1}^n [w_i \cdot f_i(x)] \quad (7)$$

$$w_i > 0, \quad \forall i \in \{1, \dots, n\} \quad f_i(x) > 0, \quad \forall i \in \{1, \dots, n\}$$

While the multiplicative scalarization technique provides a way of converging upon a single optimal solution, it may not always account for the diverse trade-offs among the objectives. The Pareto Optimal approach⁵³ is another significant technique in the field of multi-objective optimization. This method aims to find a set of solutions for which no other solutions can improve one objective without worsening at least one other objective.

Pareto optimality and tchebycheff methods

The concept of Pareto optimality is expressed mathematically as follows: given a set of solution vectors $\mathbb{X}$, a solution $x^* \in \mathbb{X}$ is Pareto optimal if there does not exist another solution $x \in \mathbb{X}$ such that $f_i(x) \geq f_i(x^*)$ for all $i \in \{1, \dots, n\}$ and $f_j(x) > f_j(x^*)$ for at least one index j .

The Tchebycheff method, on the other hand, is a scalarization method that seeks to optimize the worst-case scenario among multiple objectives. The Tchebycheff method uses a reference point z^* and the weights w_i to guide the optimization process.

Mathematically, given a solution vector x , the Tchebycheff scalarized objective function is formulated as:

$$T(x) = \max_{i=1}^n \{w_i |f_i(x) - z_i^*|\} \quad (8)$$

The objective of the Tchebycheff method is then to find a solution vector x that minimizes the scalarized objective function:

$$\min_x T(x) = \min_x \max_{i=1}^n \{w_i |f_i(x) - z_i^*|\} \quad (9)$$

It should be noted that the weight vector w in the Tchebycheff method is also normalized, meaning $\sum_{i=1}^n w_i = 1$, and each weight w_i is non-negative. The reference point z^* is usually chosen to be an ideal point where all objectives attain their optimal values. The Tchebycheff method is particularly effective when dealing with “concave” regions of the Pareto front, ensuring the generation of well-distributed Pareto optimal solutions.

Purpose of the train_flag Parameter

The `train_flag` parameter is introduced to provide flexibility in the implementation of the MOC-MAB algorithm, allowing for different operational scenarios:

- **Training Mode** (`train_flag = True`): When the `train_flag` is set to `True`, the algorithm operates in a learning mode. It updates its internal model parameters (e.g., context covariance matrices, regression parameters, and weights) based on the rewards received from user interactions. This mode is essential during the training phase, where the algorithm learns from user feedback to refine its recommendations.
- **Evaluation Mode** (`train_flag = False`): When the `train_flag` is set to `False`, the algorithm enters an evaluation mode. In this mode, the algorithm does not update its internal parameters but instead uses the learned model to generate recommendations based on previously acquired knowledge. This setting is particularly useful for:
 - *Offline evaluation*: Testing the performance of a pre-trained model on historical data without altering its learned parameters.
 - *Cold start scenarios*: Applying an initial model while gathering data for future training.
 - *Real-world deployment*: Preventing live updates to the model during certain periods, ensuring system stability and reproducibility in operational environments.

By incorporating the `train_flag`, the algorithm balances adaptability (learning from new interactions) and consistency (using a stable model for evaluation or deployment), depending on the requirements of the use case.

Differences from classical MAB approaches

The proposed MOC-MAB algorithm extends the classical MAB framework in several significant ways:

1. **Multi-Objective Optimization**: Classical MAB approaches such as e-Greedy, Upper Confidence Bound (UCB), and Thompson Sampling focus on optimizing a single objective, typically maximizing immediate rewards. In contrast, the proposed MOC-MAB algorithm incorporates multi-objective optimization, enabling it to balance multiple conflicting objectives such as relevance and fairness. This capability is particularly critical in smart tourism, where diverse user needs and business goals must be simultaneously addressed.
2. **Contextual Integration**: While classical MAB approaches do not consider context, the proposed algorithm leverages contextual information such as time, location, and user preferences to make personalized recommendations. By incorporating context, the algorithm adapts to dynamic user behaviors and environments, significantly improving the relevance of the recommendations.

3. **Adaptive Learning:** Classical MAB algorithms often operate in static environments, with limited adaptability to evolving user preferences and contexts. The MOC-MAB algorithm employs an online learning framework, allowing it to continuously learn from user feedback and refine its recommendations in real-time. This adaptability is crucial for dynamic domains such as smart tourism.
4. **Scalarization for Trade-Offs:** The proposed method uses scalarization techniques, such as multiplicative and Tchebycheff scalarization, to transform multiple objectives into a single optimization problem. This allows for effective balancing of trade-offs between objectives, a feature absent in classical MAB methods.
5. **Practical Application:** While classical MAB algorithms are primarily theoretical or applied in static datasets, the proposed approach is validated in real-world smart tourism scenarios. For example, it demonstrates the ability to recommend a mix of popular and niche tourist attractions, ensuring diversity and fairness while maintaining user satisfaction.

These differences position the proposed MOC-MAB algorithm as a robust and versatile solution for context-aware multi-objective recommendation tasks, significantly advancing the classical MAB framework.

Complexity analysis

The computational complexity of the proposed MOC-MAB algorithm is analyzed in terms of its training and inference phases.

Training Phase: During training, the algorithm updates the reward estimations for each arm based on the context vectors and scalarization weights. Let K denote the number of arms, d the dimensionality of the context vectors, and O the number of objectives. The reward computation involves matrix operations of complexity $O(d^2)$ per arm and scalarization of complexity $O(O)$. Therefore, for each training iteration, the overall complexity is $O(K \cdot (d^2 + O))$. Over T iterations, the total training complexity becomes $O(T \cdot K \cdot (d^2 + O))$.

Inference Phase: Inference involves selecting the arm with the highest scalarized reward for a given context. The reward calculation for each arm requires $O(d)$ operations, while scalarization adds $O(O)$. Hence, the per-iteration complexity for inference is $O(K \cdot (d + O))$. Over T iterations, the inference complexity is $O(T \cdot K \cdot (d + O))$.

Scalability: The algorithm scales linearly with the number of arms K , context dimensionality d , and objectives O . The scalarization and matrix updates, though computationally intensive, are efficiently implemented to handle large-scale contexts. The incremental learning approach ensures that the computational cost remains manageable in real-time scenarios.

Comparison with LinUCB: Compared to LinUCB, which has a training complexity of $O(T \cdot K \cdot d^2)$ and inference complexity of $O(T \cdot K \cdot d)$, the proposed approach adds minimal overhead due to the inclusion of multi-objective optimization. This trade-off is justified by the ability to optimize multiple conflicting objectives simultaneously.

The proposed MOC-MAB algorithm achieves a balance between computational efficiency and multi-objective optimization, making it suitable for dynamic and resource-constrained applications such as smart tourism.

Complexity of the optimization approach

The optimization process in the proposed MOC-MAB algorithm involves balancing multiple conflicting objectives through scalarization techniques. Below, we detail the complexity of these optimization steps and their implications:

1. **Scalarization Overhead:** The MOC-MAB algorithm uses scalarization methods, such as multiplicative and Tchebycheff scalarization, to combine multiple objectives into a single value for optimization. Let O denote the number of objectives. The scalarization operation incurs a complexity of $O(O)$ per arm. This overhead is minimal and scales linearly with the number of objectives, ensuring computational efficiency even with multiple conflicting goals.
2. **Contextual Vector Evaluation:** For each arm, the algorithm evaluates its reward by leveraging the context vector $x_t \in \mathbb{R}^d$. This involves matrix multiplications for reward prediction, with a complexity of $O(d^2)$ per arm due to the inverse covariance matrix updates. For K arms, the complexity is $O(K \cdot d^2)$, which is typical for contextual bandit algorithms like LinUCB.
3. **Trade-Off Management:** The algorithm manages trade-offs between objectives by optimizing the scalarized reward function. For each recommendation, the optimization process evaluates the trade-offs across all K arms, with a complexity of $O(K \cdot O)$. This ensures that the chosen action aligns with the multi-objective criteria while maintaining computational feasibility.
4. **Scalability:** The overall optimization process during training and inference involves computing scalarized rewards for all arms and updating the contextual parameters. Across T iterations, the total complexity of the optimization approach is $O(T \cdot K \cdot S(d^2 + O))$. This linear scalability with respect to arms, contexts, and objectives makes the approach suitable for large-scale real-world applications.
5. **Comparison with Other Multi-Objective Methods:** Compared to traditional single-objective MAB algorithms, the scalarization step introduces a modest additional overhead proportional to the number of objectives. However, this overhead is offset by the ability to balance conflicting objectives, making the optimization approach more versatile and practical for dynamic, multi-objective environments such as smart tourism.

The optimization complexity of the proposed approach is comparable to that of contextual bandit algorithms while incorporating additional capabilities to handle multi-objective trade-offs effectively. The algorithm

achieves a balance between computational efficiency and optimization flexibility, ensuring scalability to real-world scenarios.

Experimental evaluation
Experimental setup

Datasets

To evaluate the performance of our proposed MOC-MAB algorithm in the domain of smart tourism, we conducted experiments using two datasets: a designed dataset that simulates real-world scenarios and the TripAdvisor dataset, described in Table 2.

The real-world dataset, referred to as the “Designed Dataset,” is a curated dataset that reflects realistic smart tourism scenarios. It includes user interactions, contextual information (e.g., time, location, and weather conditions), and user feedback (e.g., ratings and satisfaction levels). This dataset is not synthetic but is derived from actual user interactions and curated to simulate the complexity and diversity of real-world tourism conditions.

In each interaction, the contextual bandits algorithm randomly selects one out of 10 possible actions (touristic places) as the recommended arm. The reward for each interaction is determined based on the tourist’s response, indicating whether they liked or disliked the recommendation. For instance, a tourist is presented with 10 potential recommendations, and one is chosen randomly as the recommended arm. This process mimics real-life scenarios where tourists are provided with multiple choices. The reward value provides crucial feedback for the algorithm to learn and improve its future recommendations.

Moreover, the dataset included 10-dimensional context vectors associated with each arm (touristic place), capturing various contextual information relevant to the tourist’s preferences and interests. By abstracting the dataset in this manner, we were able to create a realistic simulation of the smart tourism domain, enabling us to evaluate the effectiveness of our algorithm compared to baseline algorithms in providing personalized and diverse recommendations. For example, the algorithm can identify touristic places with high historical relevance, excellent food ratings, and suitability for families with children, while also considering factors like the tourist’s location and time.

The TripAdvisor dataset contains valuable insights from user reviews and ratings, reflecting their satisfaction levels, preferences, and the economic impact of various tourism offerings. By analyzing this dataset and incorporating multi-objective optimization into the contextual bandit recommender system, we can generate recommendations that not only optimize for a single objective but also consider the broader context and goals of smart tourism.

The TripAdvisor dataset used in this study is a subset of the Four-City Dataset, which originally contains 878,561 reviews (1.3 GB) from 4333 hotels crawled from TripAdvisor⁵⁴. The dataset has been filtered to include only actions that occur more than 1000 times, ensuring sufficient data for training and evaluation. Attributes such as cleanliness, location, rooms, and service quality are included, making it ideal for evaluating the performance of the MOC-MAB algorithm in generating multi-objective recommendations.

Both the Designed Dataset and the TripAdvisor Dataset were used separately in the experiments. The Designed Dataset was employed to evaluate the algorithm’s ability to adapt to a curated, diverse real-world tourism scenario with detailed contextual attributes such as time of day, weather, and user preferences. On the other hand, the TripAdvisor Dataset was used to benchmark the algorithm against a publicly available dataset containing hotel reviews and ratings, allowing for comparisons in a more traditional recommendation context. The algorithm’s performance was evaluated independently on both datasets to highlight its adaptability to different types of contexts and objectives. The MOC-MAB algorithm demonstrated strong performance on the Designed Dataset. It achieved a higher cumulative reward and lower cumulative regret compared to baseline algorithms, indicating its ability to leverage detailed contextual features like weather, user preferences, and

Dataset	Interactions	Context
Designed dataset	10,000	Tourist’s location
		Time of day
		Weather conditions
		Historical relevance of the place
		Food rating of the place
		Kid-friendly
		Nature quotient
		Adventure activities
		Shopping opportunities
TripAdvisor Dataset	436,390	Hotel’s overall ratings
		Cleanliness
		Location
		Rooms
		Service sleep quality

Table 2. Datasets.

activity suitability. This performance underscores the algorithm's adaptability in real-world tourism scenarios with multidimensional context. On the TripAdvisor Dataset, the algorithm outperformed baseline approaches in terms of both CTR and cumulative rewards. The simpler context features (e.g., hotel ratings, cleanliness, and location) in this dataset posed fewer challenges for the algorithm, resulting in faster convergence and effective exploitation of available data. These results validate the algorithm's robustness across datasets with varying complexity and contextual diversity. The comparative analysis on the two datasets highlights the algorithm's flexibility, making it suitable for both detailed real-world applications and traditional benchmark datasets.

Evaluation metrics

While MAB problems inherently operate in real-time, we leverage historical data for an evaluation approach similar to¹⁴. In our computational simulations, we take into account several benchmark models: LinUCB, Random Bandit, Thompson Sampling, and Epsilon-Greedy.

The proposed MOC-MAB algorithm is tested for varying values of α (α), where α represents the trade-off parameter in the scalarization function used to balance conflicting objectives such as relevance and fairness. A smaller α places more emphasis on fairness, while a larger α prioritizes relevance. The range of α values is chosen as 0.1–1.0 for consistency across experiments.

Figure 4 shows the mean rewards for different values of α , demonstrating that $\alpha = 0.1$ yields the highest mean reward. Figure 5 presents the click-through rate (CTR) for the same α range, further validating the trade-offs between fairness and relevance across time steps.

The benchmarks are defined as follows: The random selection algorithm, which selects an action randomly from the available options without considering any context or previous feedback, serves as a rudimentary baseline for comparison against more complex algorithms. Thompson Sampling uses a probabilistic approach that makes decisions based on posterior distribution samples, modeling the uncertainty of each arm's rewards with a Beta distribution. The algorithm naturally explores arms with uncertain reward estimates, striking a balance between exploration and exploitation. Epsilon-Greedy, a simple exploration-exploitation strategy, primarily exploits known good actions but also allows for a controlled amount of exploration to potentially discover better actions. It selects the highest estimated mean reward action with a probability of $(1 - \epsilon)$ and a random action with a probability of ϵ . The exploration-exploitation balance is regulated by the epsilon parameter. LinUCB is a linear upper confidence bound algorithm. It is a contextual bandit algorithm that uses a linear model to estimate rewards and adds an exploration bonus based on the upper confidence bound of the estimates. This approach helps in balancing the exploration and exploitation trade-off, as it tends to select actions that have a high upper confidence bound, which could either be due to a high estimated reward or high uncertainty. These benchmark algorithms offer various action selection strategies in a contextual bandit setting.

Given our approach of recommending a single touristic place at each time point, we employ clickthrough rate (ctr) as a measure of recommendation performance, where: The clickthrough rate (ctr) is calculated as follows:

$$CTR = \frac{\sum_{i=1}^N r_i}{N} \quad (10)$$

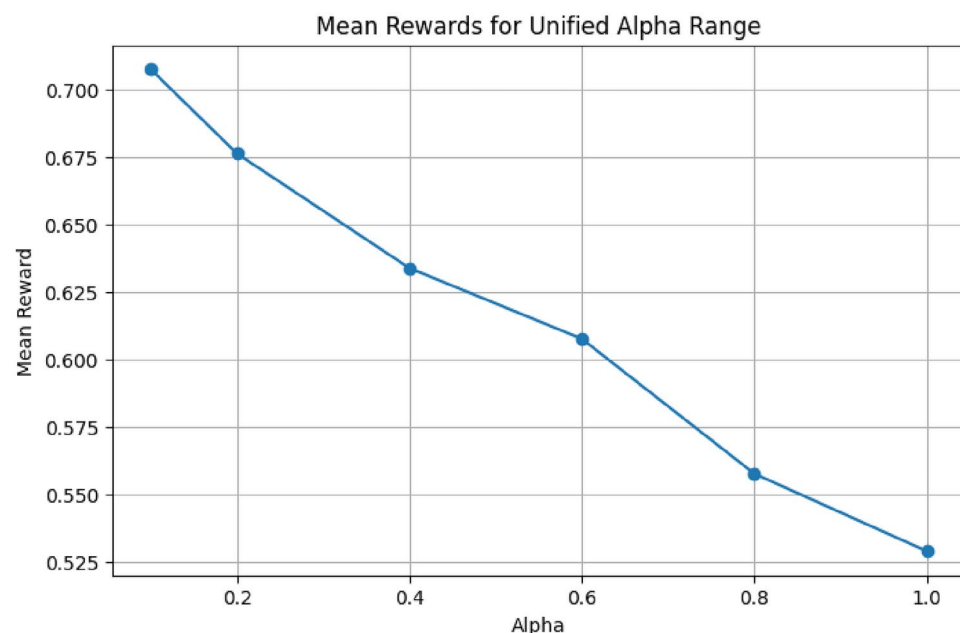


Fig. 4. Mean rewards for unified alpha range (α values: 0.1–1.0).

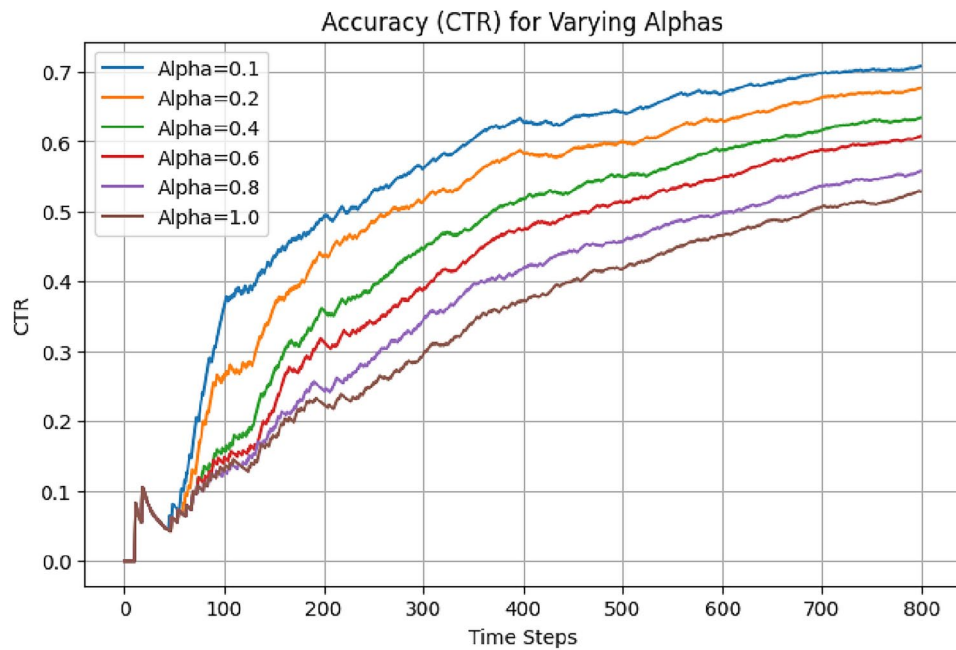


Fig. 5. Accuracy (CTR) for unified alpha range (α values: 0.1–1.0).

Where N is the total number of users and r_i is the like of the i -th user, which is 1 if the user liked the recommended place and 0 otherwise.

Reward History is an array that holds the reward obtained at each time step t . It can be represented as $R = [r_1, \dots, r_t, \dots, r_T]$ where r_t is the reward at time step t and T is the total number of rounds or time steps. Cumulative Reward History is an array that holds the cumulative sum of the rewards up to each time step. It can be represented $C_{reward} = [C_{reward_1}, \dots, C_{reward_t}, \dots, C_{reward_T}]$ where C_{reward_t} is the cumulative sum of rewards up to time step t . This can be mathematically defined as:

$$C_{reward_t} = \sum_{i=1}^t r_i \quad (11)$$

Cumulative Regret History is an array that holds the cumulative regret up to each time step. Regret at a time step t is defined as the difference between the maximum possible reward and the actual obtained reward at that time step. So, the cumulative regret can be represented $C_{regret} = [C_{regret_1}, \dots, C_{regret_t}, \dots, C_{regret_T}]$ where C_{regret_t} is the cumulative regret up to time step t . This can be mathematically defined as:

$$C_{regret_t} = \sum_{i=1}^t (regret_{\max,i} - regret_i) \quad (12)$$

where $regret_{\max,i}$ is the maximum possible reward at time step i .

In real-world scenarios, tourists are often presented with multiple recommendations, but their choice may not always strictly align with their preferences or context due to various factors such as exploration of new options, unavailability of information, or limited familiarity with alternatives. In our simulation, random selection is used during the algorithm's exploratory phase to ensure diverse options are considered and sufficient data is gathered to refine the recommendation policy.

Random selection in the experimental setup simulates an initial exploration step, enabling the algorithm to observe user reactions across a range of recommendations. This phase is crucial for algorithms like contextual bandits, which balance exploration and exploitation. Exploration helps the system learn user preferences over time, while exploitation applies this knowledge to provide tailored recommendations.

By incorporating random selection as a starting point, the algorithm can simulate the real-world behavior where tourists might initially explore diverse options before settling on choices that align closely with their preferences. As the algorithm learns from user feedback, it transitions to recommending items that are increasingly personalized and aligned with the user's tastes and contexts.

All experiments were conducted using the Google Colaboratory (Google Colab) environment, which provides an online platform for executing Python code with access to GPU and TPU accelerators. Google Colab offers a Jupyter Notebook interface, enabling seamless integration of code, data visualization, and experimentation. For this study, we utilized the Python programming language (version 3.9) and essential libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn for data manipulation, computation, and visualization. The experiments

were executed on a virtual machine with 12.72 GB of available RAM and GPU acceleration enabled, ensuring efficient processing of the datasets and the proposed algorithm.

Evaluation results

Cumulative regret is a widely recognized metric in contextual bandit algorithms for evaluating performance. It represents the cumulative difference between the reward obtained by the algorithm and the reward that would have been achieved by consistently selecting the optimal action at each time step. This metric quantifies the learning efficiency of the algorithm, offering insight into how effectively it balances exploration (gathering new information) and exploitation (leveraging known information). A lower cumulative regret indicates that the algorithm is effectively learning from its exploration, progressively optimizing its decisions, and successfully integrating contextual information to enhance recommendations over time. Figure 6 compares the cumulative regret of the proposed Multi-Objective Contextual MAB (MOC-MAB) algorithm against several baseline methods, including LinUCB, Random Selection, Thompson Sampling, and Epsilon-Greedy.

The results demonstrate that the proposed MOC-MAB algorithm consistently achieves lower cumulative regret compared to all baselines, including LinUCB. This indicates that MOC-MAB learns more effectively, adapting to contextual information and efficiently balancing exploration and exploitation. In contrast, LinUCB exhibits higher cumulative regret, reflecting its limited capacity to address the multi-objective nature of the problem. Additionally, heuristic approaches like Random Selection, Epsilon-Greedy, and Thompson Sampling fail to capture the contextual complexity of the environment, leading to significantly higher regrets. Overall, the results validate that MOC-MAB not only minimizes opportunity costs but also outperforms competing methods in achieving faster and more effective decision-making.

Compared to baseline algorithms, namely LinUCB, random selection, Thompson sampling, and epsilon-greedy, our proposed Multi-Objective Contextual MAB algorithm offers distinct advantages. The scalarization approach allows accounting for the relative importance of different context elements, resulting in more tailored and accurate recommendations. The algorithm's ability to optimize multiple objectives simultaneously enables it to strike a balance between the diversity of smart city touristic places and tourists' preferences. By considering these multiple objectives, our algorithm outperforms baseline approaches, providing a more comprehensive and personalized recommendation system that enhances the tourist experience.

To evaluate the performance in terms of cumulative rewards, we added a comparative analysis as illustrated in Fig. 7. The Multi-Objective Contextual MAB achieves cumulative rewards comparable to LinUCB while outperforming all baseline algorithms. This demonstrates the robustness of our proposed approach in balancing exploration and exploitation and optimizing for multiple objectives simultaneously.

The optimal value of α for the Multi-Objective Contextual MAB (MOC-MAB) algorithm using Pareto Optimal, Tchebycheff, and Weighted-Multi scalarization methods was consistent at $\alpha = 1.0$. This consistency indicates the robustness of the scalarization strategies in managing multiple objectives effectively. The analysis highlights that Weighted-Multi scalarization outperforms Pareto Optimal and Tchebycheff methods in terms of both cumulative rewards and computational efficiency.

By analyzing the CTR values, we can gain insights into how effective the recommendation system is in generating recommendations that are preferred by users. A higher CTR indicates that a larger proportion of

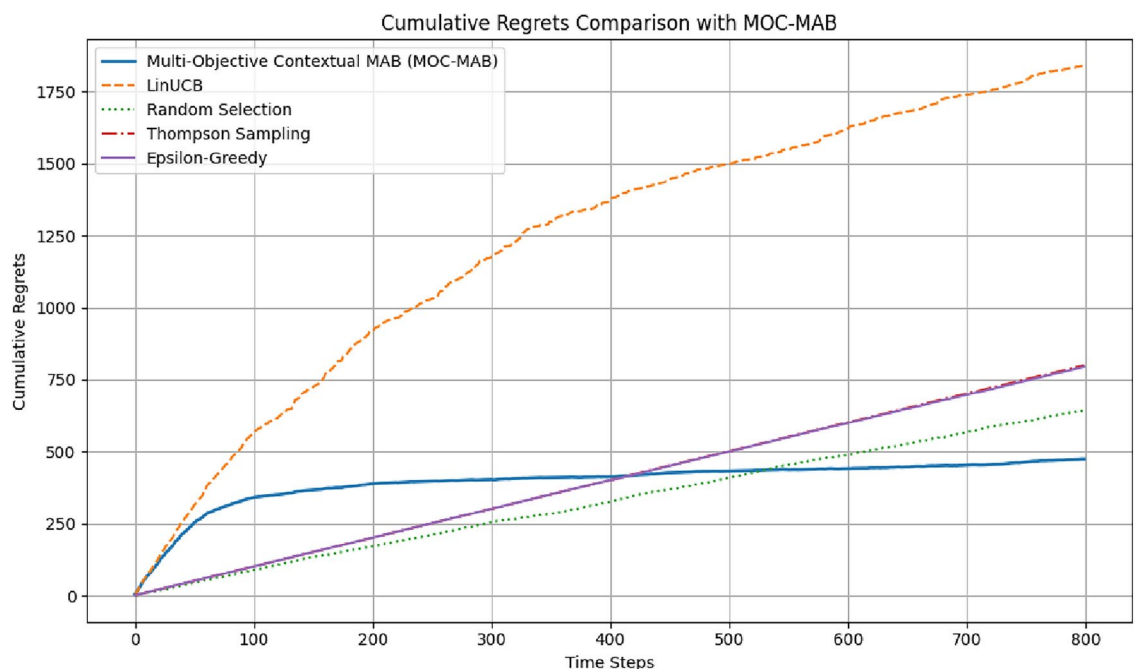


Fig. 6. Commutative regrets comparison.

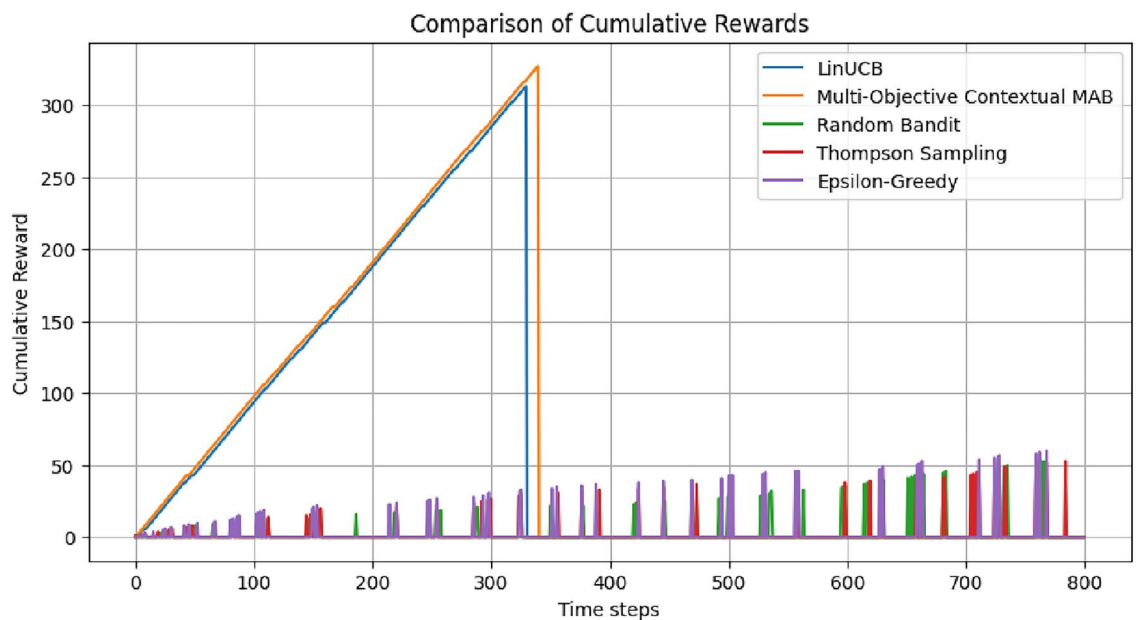


Fig. 7. Comparison of cumulative rewards for Multi-Objective Contextual MAB and baseline algorithms.

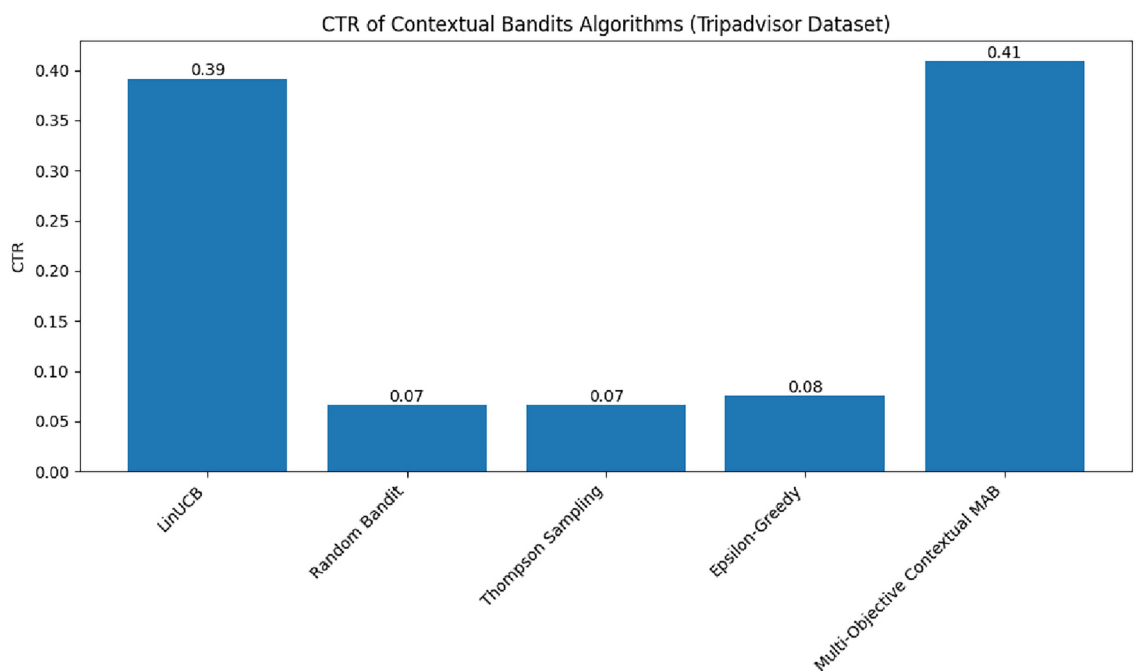


Fig. 8. CTR Comparison for contextual bandit algorithms.

users are interacting positively with the recommended touristic places, while a lower CTR may suggest a need for improvement in the recommendation strategy. CTR provides a quantitative measure of user engagement and satisfaction with the recommended touristic places and helps evaluate the performance of the recommendation system. In Fig. 8, the CTR of the proposed Multi-Objective Contextual MAB outperforms the baselines. The superior performance of the proposed algorithm indicates its potential to provide better personalized recommendations and capture user preferences more accurately.

Figure 9 presents the CTR values over time based on the history of rewards. The Multi-Objective Contextual MAB using Weighted-Multi exhibits an increasing CTR, indicating its ability to capture user preferences and provide more relevant recommendations. In contrast, the CTR of Multi-Objective Contextual MAB using Pareto Optimal and using Tchebycheff show a decreasing trend. These algorithms may have struggled to effectively

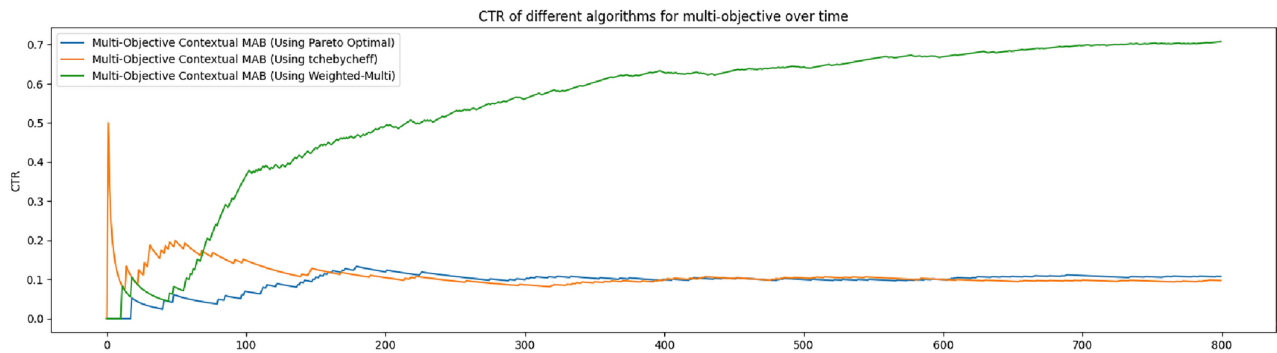


Fig. 9. CTR of different algorithms for multi-objective over time.

balance multiple objectives or address diverse user preferences, resulting in less appealing recommendations. The findings emphasize the importance of algorithm selection and design in multi-objective recommender systems.

The execution time of the Multi-Objective Contextual MAB (Using Weighted-Multi) algorithm is observed to decrease compared to the other two algorithms, as shown in Fig. 9. This reduction in execution time can be attributed to the efficient computation and decision-making process employed by the Weighted-Multi approach. By incorporating weights to balance multiple objectives, the algorithm can quickly evaluate and select the most promising actions for recommendation, resulting in faster execution. This improved efficiency allows for more responsive and real-time recommendations, enhancing the overall user experience and system performance.

The Multi-Objective Contextual MAB (Using Weighted-Multi) algorithm exhibits superior performance compared to the other two algorithms, namely Multi-Objective Contextual MAB (Using Pareto Optimal) and Multi-Objective Contextual MAB (Using Tchebycheff). This is evident from the observed increase in cumulative rewards and decreases in cumulative regrets over time presented in Figs. 6 and 7. The Weighted-Multi approach utilizes a weighted scalarization technique to balance multiple objectives, allowing for effective exploration and exploitation of the recommendation space. The Weighted-Multi algorithm demonstrates faster convergence and better performance in a shorter number of rounds compared to the other algorithms. This efficiency can be attributed to its ability to quickly identify and select the most promising actions based on the weighted objectives. As a result, users receive more accurate and relevant recommendations in a shorter period, enhancing their overall satisfaction and improving the system's performance.

To evaluate the performance of the proposed Multi-Objective Contextual Multi-Armed Bandit (MOC-MAB) algorithm, we focus on multiple conflicting objectives: relevance, fairness, diversity, novelty, and serendipity. To balance relevance and fairness, the proposed Multi-Objective Contextual MAB (MOC-MAB) algorithm incorporates a scalarization framework that assigns weights to different objectives. This enables dynamic adjustment of trade-offs based on the application context. For instance, in scenarios emphasizing user satisfaction, higher weights can be assigned to relevance, while fairness can be prioritized in settings requiring equitable distribution. The weights are tuned using grid search to identify optimal parameter values, as demonstrated by the results for varying α values in Figs. 4 and 5. This approach ensures the algorithm adapts to diverse contexts and achieves a balance between conflicting objectives like relevance and fairness, enhancing the overall quality of recommendations.

- Relevance measures how well a recommendation aligns with the user's preferences. It is derived directly from the reward received at each time step:

$$R_t = r_t$$

where r_t is the reward at time t .

- Fairness ensures equitable distribution of recommendations across users or items. It is computed as:

$$F_t = \frac{1}{1 + \text{abs}(a_t - a_{\text{optimal}})}$$

where a_t is the selected action, and a_{optimal} is the optimal action.

- Diversity measures the variety in recommendations. It is computed as the variance of the rewards over time:

$$D_t = \text{Variance}(R_{1:t})$$

- Novelty rewards recommendations that introduce new or less familiar items. It is defined as:

$$N_t = \frac{1}{\text{Popularity}(a_t)}$$

where $\text{Popularity}(a_t)$ represents the popularity of the recommended action.

- Serendipity measures the ability to recommend surprising yet relevant items. It is calculated as:

$$S_t = \frac{1}{1 + e^{-\text{abs}(R_t - a_{\text{optimal}})}}$$

Figures 10, 11, 12, 13 and 14 present the cumulative rewards achieved by the proposed MOC-MAB algorithm and baseline methods (LinUCB, Random Selection, Thompson Sampling, and Epsilon-Greedy) for each metric.

Figure 10 shows that MOC-MAB achieves higher cumulative rewards for relevance compared to baselines. This demonstrates its ability to recommend items that closely align with user preferences over time.

Figure 11 illustrates that MOC-MAB maintains higher fairness scores, distributing recommendations equitably across users. This highlights its ability to balance fairness with other objectives.

Figure 12 indicates that MOC-MAB performs better in diversity compared to other methods, ensuring a wider range of recommendations that cater to varied user preferences.

Figure 13 demonstrates MOC-MAB's ability to recommend novel items effectively, outperforming baselines by encouraging exploration of less familiar options.

Figure 14 highlights that MOC-MAB excels in recommending serendipitous items, effectively balancing relevance and surprise.

The proposed MOC-MAB algorithm consistently outperforms baseline methods across all metrics. The results validate its ability to balance conflicting objectives like relevance, fairness, diversity, novelty, and serendipity. Compared to single-objective methods, MOC-MAB dynamically integrates these objectives, ensuring recommendations are both personalized and diverse. This multi-objective optimization capability is particularly advantageous in real-world applications like smart tourism, where user satisfaction depends on a balance of varied factors. By leveraging contextual information, MOC-MAB effectively balances exploration and exploitation, achieving superior performance in cumulative rewards and minimizing cumulative regret across metrics. The findings demonstrate that MOC-MAB provides a robust and scalable solution for recommendation systems, ensuring high-quality, user-centered recommendations that cater to diverse preferences and contexts.

By adopting a multi-objective approach in the contextual bandit recommender system for smart tourism, we can provide more comprehensive and personalized recommendations that enhance the overall travel experience, support the growth of tourism destinations, and foster a vibrant and diverse tourism ecosystem. The proposed MOC-MAB algorithm demonstrates significant advantages in leveraging contextual information, such as user demographics, preferences, time, and location, to provide personalized recommendations. Unlike traditional methods, which often overlook these factors, MOC-MAB dynamically integrates this contextual data to adapt its recommendations in real time. This adaptability ensures that users receive more relevant and engaging suggestions, enhancing the overall quality of recommendations. For instance, MOC-MAB can recommend time-

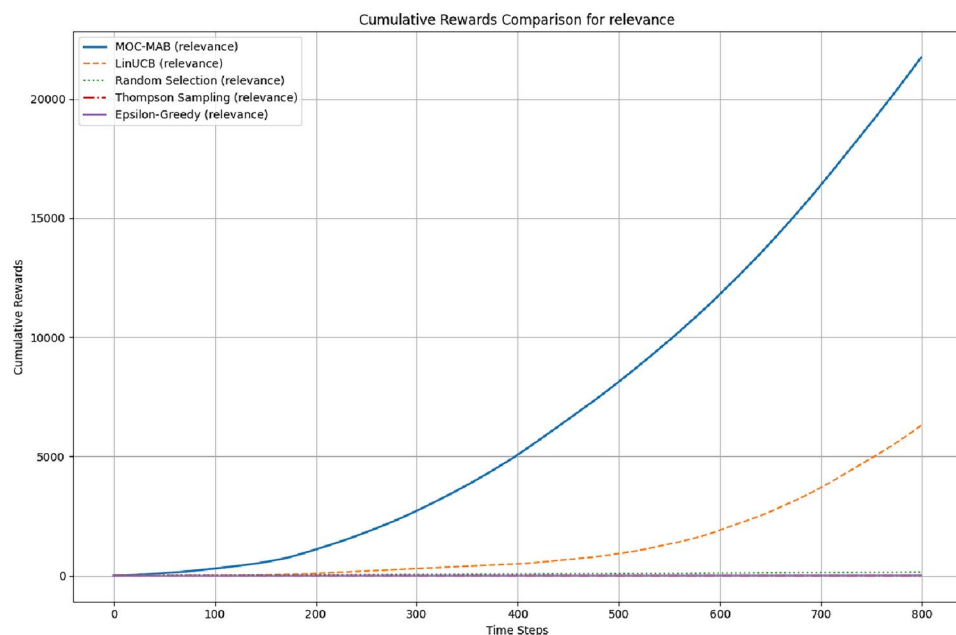


Fig. 10. Cumulative rewards for relevance.

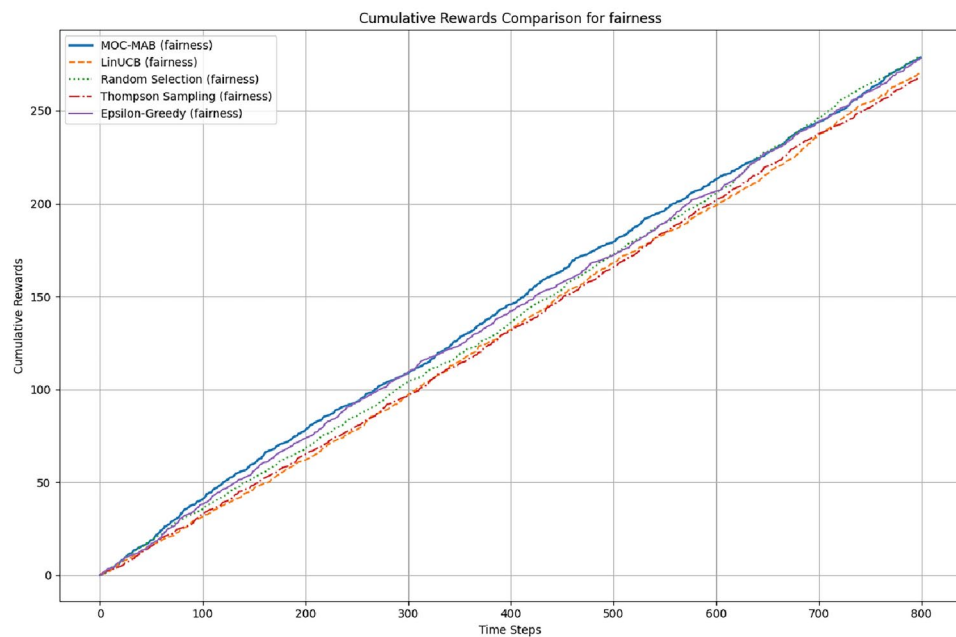


Fig. 11. Cumulative rewards for fairness.

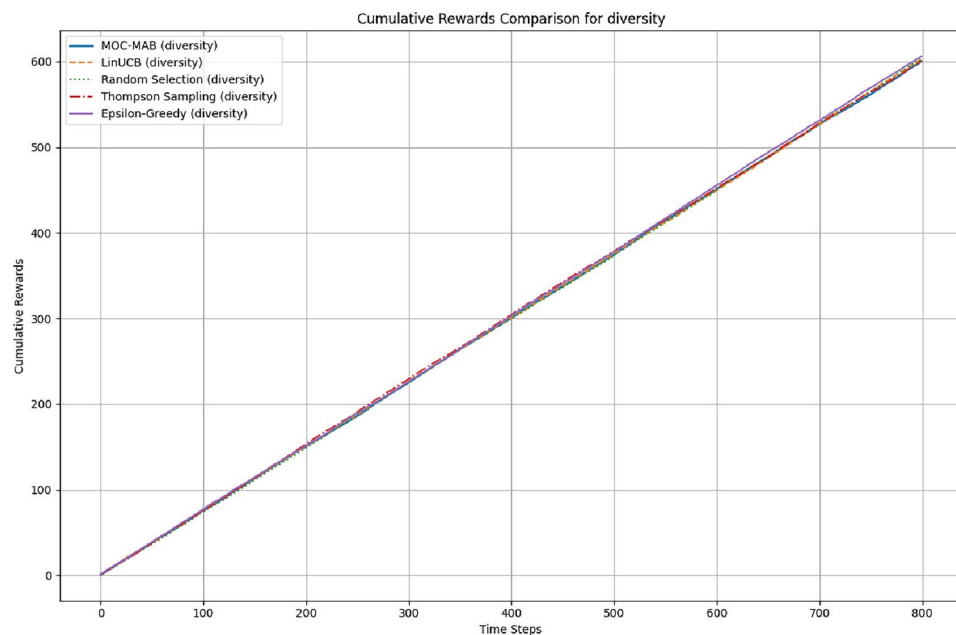


Fig. 12. Cumulative rewards for diversity.

sensitive activities, such as evening entertainment options or nearby attractions during travel breaks, tailoring its recommendations to the specific context of each user.

Conclusion and future work

Context-aware recommender systems

Recommender systems have traditionally focused on analyzing user-item interactions to generate personalized recommendations. However, these approaches often overlook important contextual details such as time, environmental conditions, and other relevant factors. Context-aware recommender systems offer a solution by incorporating various contextual elements to enhance the recommendation process. By considering factors like user profiles, temporal dynamics, and environmental conditions, these systems gain a deeper understanding of user preferences and behavior. This allows for more personalized and relevant recommendations that adapt to

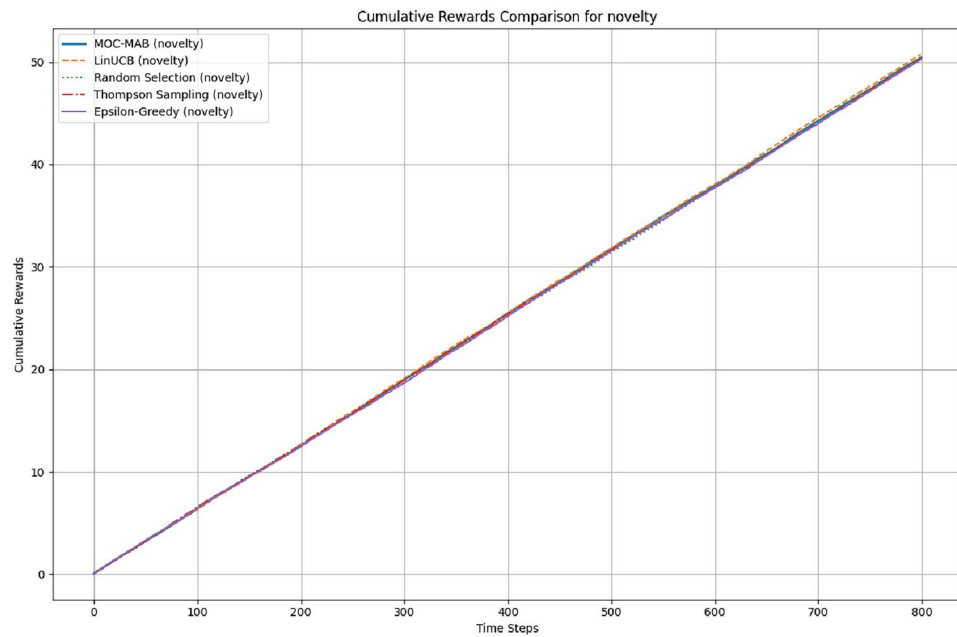


Fig. 13. Cumulative rewards for novelty.

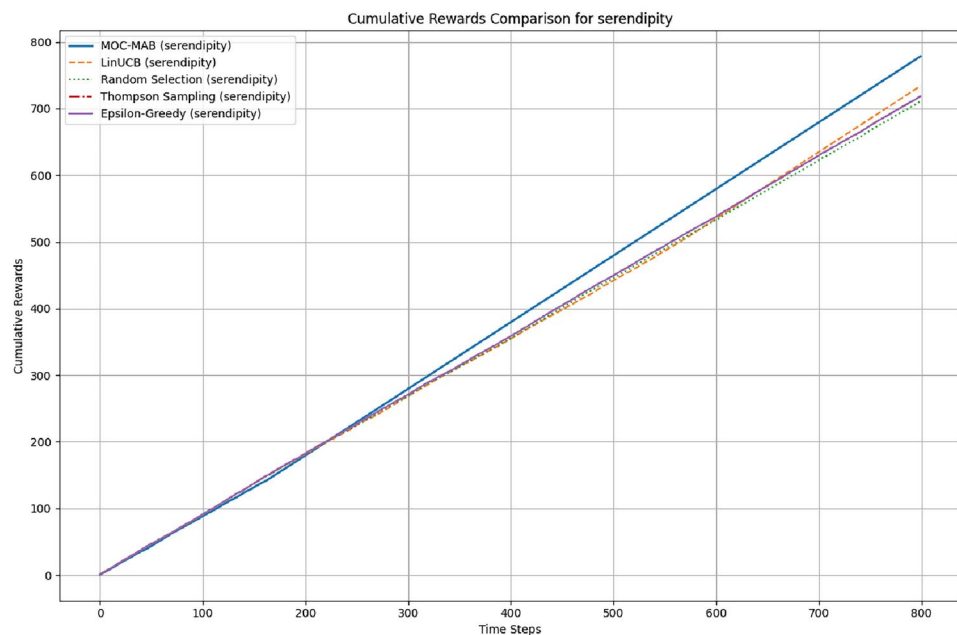


Fig. 14. Cumulative rewards for serendipity.

the specific needs of each user. Contextual information empowers recommender systems to adapt to changing circumstances, ultimately improving the overall user experience and satisfaction. Despite the progress made in context-aware recommender systems, their approaches often rely on static models that struggle to learn and adapt to ongoing user interactions. This limitation becomes more pronounced as user preferences and contextual factors evolve over time. However, reinforcement learning, particularly in the context of the Multi-Armed Bandit (MAB) problem, offers a promising solution. MAB algorithms provide a dynamic and adaptive framework for context-aware recommender systems, allowing for better performance by capturing the evolving nature of user behavior and context.

Multi-objective optimization in contextual bandit algorithms

While contextual bandit algorithms are effective in optimizing a single objective, they often neglect important factors like diversity and fairness in recommendation systems. To address this, researchers have started exploring the integration of multi-objective optimization techniques into contextual bandit algorithms. Multi-objective optimization aims to find a balance between conflicting objectives, such as relevance, diversity, fairness, novelty, and serendipity. By incorporating multi-objective optimization, contextual bandit algorithms can consider and optimize for these diverse objectives, leading to more comprehensive and personalized recommendations.

Proposed approach and key findings

This paper presents a new approach to address the challenges of multi-objective contextual recommender systems in the context of smart tourism in smart cities. Our proposed approach combines the advantages of contextual bandit algorithms and multi-objective optimization to create a multi-objective contextual multi-arm recommender system. The algorithm can adapt and learn from user feedback while simultaneously taking multiple objectives into account, thus embedding contextual information into the recommendation process.

The experimental results demonstrate several key findings. The proposed algorithm significantly improves cumulative rewards and click-through rates compared to baseline methods, highlighting its effectiveness in delivering personalized and relevant recommendations. Additionally, the algorithm achieves a balanced optimization of conflicting objectives such as relevance and fairness, providing diverse and equitable recommendations. These results underline the practical applicability of the Multi-Objective Contextual MAB algorithm in real-world scenarios, particularly in dynamic and multi-objective environments such as smart tourism. The bounded and fixed regret observed in the analysis further confirms the stability and efficiency of the approach, even in complex and evolving contexts.

Future research directions

The presented model and findings offer valuable insights into the design of a recommender system that combines contextual bandit algorithms and multi-objective optimization. However, there are several future research directions to be explored. Firstly, it is important to investigate the integration of user sentiments and emotions into the recommendation process. This consideration would enable the algorithm to deliver recommendations that resonate emotionally with users, enhancing their engagement and satisfaction. Secondly, conducting tests of the MOC-MAB algorithm on diverse datasets from various domains, such as healthcare, e-commerce, and education, to comprehensively assess its adaptability and effectiveness. Additionally, expanding the number of arms in the recommender system would offer users a broader range of recommendation options, catering to their individual preferences and interests. Lastly, incorporating online evaluations that capture real-time user interactions and preferences would enable the algorithm to adapt to dynamic user behaviors and evolving contextual factors more effectively. Exploring these future research directions would contribute to the advancement of recommender systems, ultimately improving the personalized recommendation experience for users.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

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